Analytics & Forecasting The Role of Big Data Analytics in Improving Demand Forecasting Accuracy

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1. Introduction

1.1 Background of the Study

In the current competitive business world, organizations have a massive challenge in terms of forecasting consumer demand accurately. The difficulty with modern markets and their volatility is that traditional forecasting models, which use historical sales data or simple statistical methods, frequently fail. The emergence of the so-called Big Data Analytics (BDA) has transformed the demand forecasting process, as it allows companies to process large, heterogeneous, and real-time data sets provided by various sources, including social media, sensors, supply chains, and customer transactions. Through predictive and prescriptive analytics, companies will be able to gain better insights into the behavior of customers, seasons, and market trends that will cause more proactive and accurate decision-making (Arguelles and Polkowski, 2023). The adoption of big data systems and technologies, including Hadoop, Spark, and machine learning algorithms, has increased the accuracy and reactiveness of demand forecasting systems in a range of industries, such as retail, manufacturing, and logistics.

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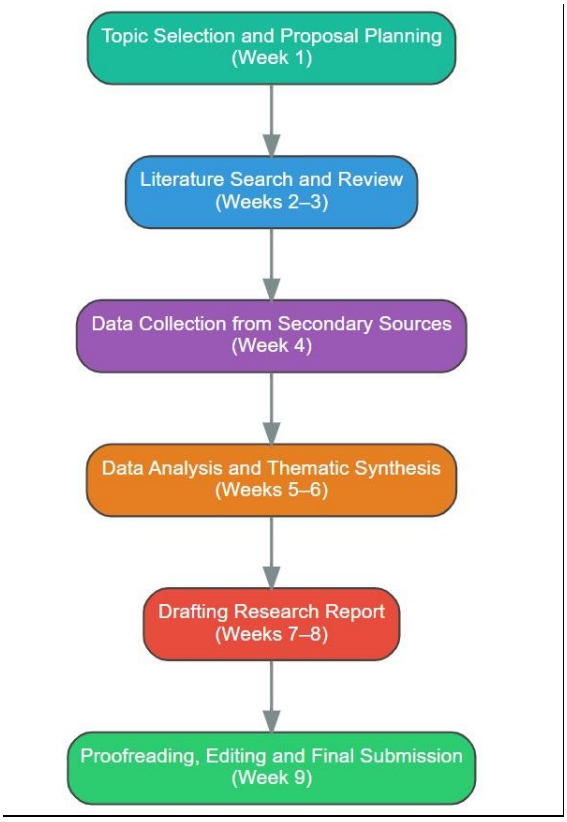

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